Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SIXTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
AUGUST 2021
SUBJECT: EMBEDDED SYSTEM DESIGN (ECE - 4053)

TIME: 2 HOURS **MAX. MARKS: 40**

Instructions to candidates
- Answer any FOUR FULL questions.
- Missing data may be suitably assumed.

1A. Explain I2C protocol. List and explain the steps for data transfer in I2C interface with relevant diagrams.

1B. What is a watchdog timer? Explain its working with relevant diagram.

(6+4)

2A. Explain round robin scheduling with an example. Explain its advantages and disadvantages.

2B. List and explain the factors on which interrupt latency depends on.

(6+4)

3A. Describe the 3 processor technologies with relevant diagrams.

3B. Compare function que scheduling with RTOS method of scheduling.

(6+4)

4A. What is a process/task? List the goals of a process/task. With a neat detailed diagram, explain the states of a task showing how tasks change the state.

4B.
```
int add (int a, int b) {    return a+b;    }
int mul (int a, int b) {    return a*b;    }
int sub (int a, int b)  {    return a-b;    }
int div (int a, int b)  {    return a/b;    }
```
Declare a function pointer which can be used to point to above functions. Assign all these functions in all possible ways to this function pointer. Write statements to call these functions in all possible ways using function pointer.

(6+4)

5A. Write main program code in C to implement a Que data structure using dynamic memory allocation. Also write code for function declarations mentioned below. Main program can have more than one Que at a given point of time. Let the node contain a single integer data. Print the data when De-queing and return -1 when trying to de-que an empty que.

```
struct node* create_Q();
EnQ (int x, struct* Que);
int DeQ ( struct* Que );
```

5B. Define a reentrant function. Explain with an example, the issue that we may encounter while using a non-reentrant function.

(6+4)

6A. Explain with relevant graphs, the time to market design metric. What is its importance.

6B. With a neat diagram depicting memory while program is running, explain the categories of program memory.

(5+5)